

PROFESSIONAL SUMMARY

AI Engineer focused on building production-ready GenAI and Agentic AI systems using FastAPI, LangChain, vector databases, and multimodal ML pipelines. Experienced in deploying scalable RAG applications, medical imaging AI systems, and real-time inference APIs for practical business and healthcare use cases.

EDUCATION

- **Indian Institute of Information Technology Lucknow** Lucknow, India
M.Sc. in Artificial Intelligence & Machine Learning; CGPA: 8.9 Aug 2024 – Present
- **RTMNU** Nagpur, India
B.Sc. in Statistics, Computer Science Mathematics; CGPA: 6.8 Aug 2018 – June 2021

EXPERIENCE

- **Blu Parrot – AI Engineer Intern** Feb 2026 – March 2026

Agri Advisory Platform:

- Architected and deployed an end-to-end AI advisory platform integrating soil (NPK), climate, and regional datasets to generate crop and fertilizer recommendations across multiple Indian states
- Built OCR-based ingestion pipelines for farmer soil reports with fallback handling for incomplete data, improving robustness and coverage across real-world inputs
- Designed a hybrid decision system combining rule-based agronomic logic with ML models, enabling context-aware recommendations at scale
- Developed scalable backend services and relational schemas supporting multi-region data onboarding and extensibility for new crops and datasets

- **Sneh Bharat Pvt. Ltd. (Mactix Global Solution) – AI Intern** Jan 2026 – Feb 2026

Medical Imaging AI Pipeline:

- Built an end-to-end medical imaging pipeline for Chest X-ray analysis, supporting DICOM and multi-format report ingestion with metadata normalization
- Integrated CheXNet (DenseNet-121) for multi-label classification (14 diseases) and automated radiology-style report generation, reducing manual diagnostic effort
- Deployed model inference via FastAPI-based REST services, enabling integration with hospital systems and real-time clinical workflows

AI Travel Assistant Backend:

- Developed a production-grade RAG pipeline using LangChain and ChromaDB, powering a conversational AI system for travel query resolution
- Designed REST APIs integrated with a React + TypeScript frontend, enabling scalable user interaction and session-based responses
- Optimized vector retrieval and ranking pipelines, improving response relevance and downstream lead conversion performance

PROJECTS

- **Tumor-Attentive GAN for Medical Image Synthesis (CE-MRI)** | *PyTorch, GANs, Medical Imaging* 
 - Designed and implemented a tumor-attentive GAN pipeline for contrast-free MRI synthesis, processing large-scale medical datasets and improving clinical feasibility
 - Integrated segmentation-guided spatial attention and custom perceptual loss functions, improving tumor-region reconstruction fidelity by 15% over baseline models
 - Achieved 28.93 dB PSNR and 0.744 SSIM, demonstrating high-quality image synthesis for diagnostic use cases
 - Built an end-to-end pipeline (preprocessing → training → evaluation → inference) with a deployed Streamlit interface for real-time clinical validation
- **Multi-Modal Fake News Detection using HAMMER++** | *PyTorch, BERT, ViLT, ViT* 
 - Built a multi-modal fake news detection system combining BERT (text) and ViT (image) representations across 10,000+ samples
 - Built a multimodal AI pipeline combining text and image embeddings for robust fake content detection, improving performance by 12% over single-modality baselines
 - Achieved F1: 0.81, AUC: 0.91, and 80.8% accuracy with optimized training pipelines using W&B, LR scheduling, and early stopping
- **Agentic RAG System with Hierarchical Multi-Agent Workflow** | *LangChain, ChromaDB, Streamlit* 
 - Developed a hierarchical multi-agent RAG system over 50+ research papers with dynamic tool usage and reasoning workflows
 - Achieved 92.4% retrieval precision@5 and 84.7% keyword accuracy through optimized embedding and retrieval pipelines
 - Deployed a full-stack Streamlit application supporting document ingestion, semantic search, and local LLM inference (Ollama/LLaMA)

TECHNICAL SKILLS

AI Systems & LLMs: RAG, LangChain, LangGraph, LoRA, QLoRA, ChromaDB, FAISS, Pinecone

Deep Learning: GANs, CNNs, U-Net, Vision Transformers (ViT), BERT, ViLT

ML Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers

MLOps & Deployment: Docker, MLflow, DVC, W&B, FastAPI, Flask, Streamlit, CI/CD

Data & Engineering: DICOM/NIfTI Processing, NiBabel, Feature Engineering, PostgreSQL, MongoDB

Languages: Python, SQL, C (Basic)

CERTIFICATIONS & ACCOMPLISHMENTS

- **IIT JAM 2024:** Secured All India Rank **320** in Statistics
- **IIT Guwahati:** Summer Project 2025 – Solving Inventory Inefficiencies Using Advanced SQL Analytics [\[Certificate\]](#)
- **ISRO Bharatiya Antariksh Hackathon 2025:** Successfully submitted idea for Government Schemes Chatbot [\[Certificate\]](#)